

Layout-Aware Translation Benchmark

A deterministic two-column document with a structured table

1. Introduction

Document translation quality depends on reliable reading order. The left column must remain separate from the right column, while headings and captions should provide context to the translation model.

2. Method

The standard pipeline parses native PDF text, detects semantic layout regions, recognizes table structure, and applies OCR only when required.

3. Results

Stable bounding boxes let LingoPane place translated text over the original page. A versioned document contract keeps the React interface independent from Docling internals.

Table 1. Expected pipeline responsibilities

Stage	Responsibility
PDF parser	Native text and coordinates
Layout	Semantic regions and order
Table	Rows, columns, and cells
oMLX	Local text translation

2.1 Architecture Across Both Columns

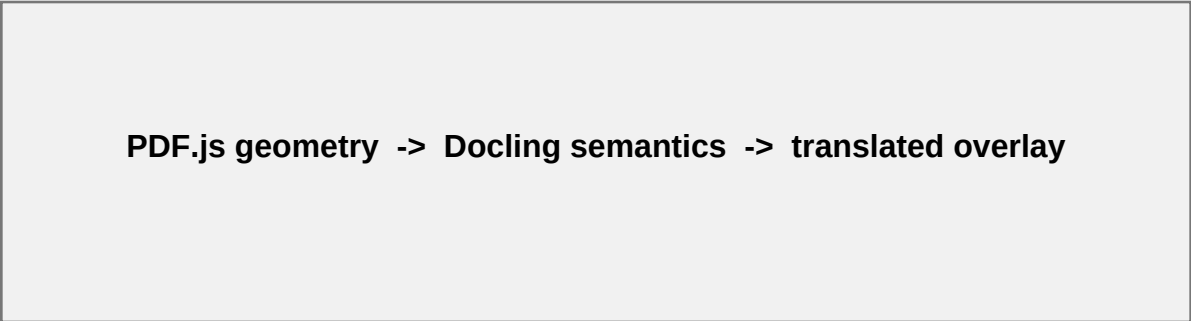
This full-width section heading intentionally crosses the column boundary and must remain ahead of both body columns.

2.1.1 Source Pane

Native PDF coordinates define the source geometry. Crop boxes and page rotation are normalized through the PDF viewport matrix.

2.1.2 Translation Pane

Docling semantic blocks are aligned back to matching PDF text runs before translated overlays and source masks are rendered.



PDF.js geometry -> Docling semantics -> translated overlay

Figure 1. A cross-column figure with a centered caption

2.2 Contributions

This paragraph is deliberately long enough to exercise translated-text fitting. A verbose target language should reach the minimum font size and expose the complete-text reader instead of silently clipping important content.